Lec. 9. Conservative dentistry

Class I amalgam cavity

1. **Outline form:** is the shape of the cavity which the Cavosurface line angle of the cavity assumes after preparation.

Access:

- Gain initial access via the most carious part of the tooth.
- Margins should be placed on sound tooth structures.
- In cl I all occlusal fissures and at least that in the developmental grooves have been included in the preparation even hen caries has not extended through out the fissure because it has been noted that carious dentin although not evident visually or radiographically seen at the base of fissures. There is a strong evidence that carious dentine may be present at the base of a sealed fissure,

(**extension *for prevention***).

2. **Retention form:** is the shape of the cavity that permits the restoration to resist displacement through the tipping or lifting force.

To provide retention the cavity have the following:

Opposing wall of should be parallel to each other or converge occlusally (50) this convergence done on buccal and lingual wall fig 7.

The floor of the cavity should be flat to prevent restoration movement.

Outline form should be small as possible to prevent displacing force on it.

3. Resistance form

Is the shape of the cavity that enables both the tooth and restoration to withstand occlusal forces without fracture. And this includes:

1. Prevention of fracture of the tooth

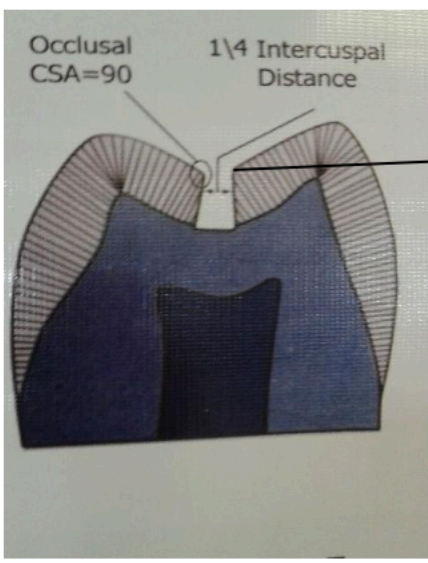
- The facio-lingual width of the preparation should not exceed 1\4 intercuspal distance fig 6. Cavo-surface line angle should be 90 – 110 Fig 6: resistance means of CL I.

- Removal of unsupported enamel by making the margin (90
-110) because less than 900 the tooth is more subjected to fracture.

- Smooth pulpal floor to prevent stress concentration area.

- Mesial & distal walls of the cavity should be parallel and slightly diverge occlusally to be within the enamel rod direction and prevent any unsupported enamel at the marginal ridge.

- All internal line angle should be rounded to prevent stress concentration area.
Facio-lingual walls converge occlusally Rounded internal line angle.



Cavo-surface line angle should be 90^0- 110^0

Preventing fracture of restoration

- The margins or Cavosurface line angle should be (90 -110) if more than this lead to fracture of restoration.
- Facio-lingual width should be 1\4 intercuspal distance because large surface

area exposed to more force and fracture occur.
- Occlusal amalgam should have thickness of (1.5-2.0 mm) to resist fracture during function.
- The pulpal floor should be smooth to prevent concentration area on restoration.

4. Convenience form:

Is the shape of the cavity that allows an adequate observation, accessability this achieved by giving good depth (1.5-2mm) and width (1\4) intercuspal distance.

5. Removal of remaining caries

Deep dentinal caries can be removed by using spoon excavator or large round bur with slow speed hand piece.

6. Finish enamel walls

Involve making the wall smooth and removing of unsupported enamel.

7. Clean the preparation:

Removal of all debris by washing the cavity and drying it.